Experiment Title: Practice of Basic Linux Terminal Commands

Topic: File and Directory Navigation in Linux Terminal

1. Objective

The objective of this lab is to learn how to use the Linux terminal for daily file and directory tasks. By completing this experiment, we aim to:

- Learn who is currently logged in using **`whoami`** and find our current location using **`pwd`**.
- Move between folders easily using **`cd`** and return home using **`cd ~`**.
- View regular files, hidden files, and file details (like size and permissions) using **`ls`** and its flags (**`-l`**, **`-a`**, **`-lh`**, **`-la`**).

2. Equipment / Software Requirements

Item	Details	Item	Details
Operating System	Linux (NixOS 26.11, Kernel 6.12)	**Host Machine**	`firefly`
User Account	`ratul`	**Command Shell**	GNU Bash 5.2 / Zsh
Terminal Emulator	Linux Terminal (Kitty / Ghostty)	**Core Tools**	GNU Coreutils (`ls`, `pwd`, `whoami`)

3. Theoretical Background

In Linux, all files and folders (called directories) are organized in a single upside-down tree structure. The very top folder is called the root folder, written as **`/`**.

When a user logs in, they start in their personal home directory (`/home/ratul`), which is also written as a tilde (`~`). The terminal allows us to move around this tree and manage files using simple text commands.

3.1 Commands Practiced in This Lab

Command	Syntax	Simple Meaning / Purpose	Example
`whoami`	`whoami`	Prints the username of the logged-in user.	`whoami`
`pwd`	`pwd`	Prints the full path of the current folder you are inside.	`pwd`
`cd`	`cd [folder]`	Moves into the specified folder.	`cd lab_demo`
`cd ~`	`cd ~`	Takes you straight back to your home folder.	`cd ~`
`ls`	`ls`	Lists the visible files and folders in the current folder.	`ls`
`ls -l`	`ls -l`	Long listing: shows file details (permissions, size, date).	`ls -l`
`ls -a`	`ls -a`	Shows *all* files, including hidden files starting with a dot (`.`).	`ls -a`
`ls -lh`	`ls -lh`	Shows file details with easy-to-read sizes (K, M, G).	`ls -lh`
`ls -la`	`ls -la`	Shows all files (including hidden ones) with full details.	`ls -la`

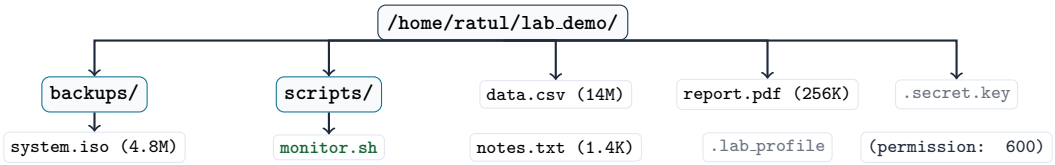
3.2 Understanding File Details (`ls -l`)

When we run **`ls -l`**, Linux shows seven pieces of information for each file:

Field	Simple Explanation
1. File Type & Permissions	10 letters (e.g., `-rw-r--r--`): the first letter shows if it is a normal file (`-`) or a directory (`d`). The next 9 letters show read (`r`), write (`w`), and execute (`x`) rights for owner, group, and others.
2. Links	How many directory links point to this file.
3. Owner & Group	Who owns the file (`ratul`) and which group it belongs to (`users`).
4. File Size	How big the file is in bytes (or in KB/MB when using `-h`).
5. Date & Time	The day and time the file was last edited.
6. File Name	The name of the file or folder.

4. Equipment / Demo Directory Setup

To practice these commands, we set up a demo folder called **lab_demo** inside /home/ratul. It contains subfolders, regular documents, a script, and hidden files:



5. Procedure / Steps

We ran the commands in order through the following simple steps:

1. Checked current user with **whoami** and checked current folder with **pwd**.
2. Listed folders inside our home directory using **ls**.
3. Entered the demo directory using **cd lab_demo** and confirmed the new location with **pwd**.
4. Listed regular files using **ls**.
5. Listed all files including hidden files using **ls -a**.
6. Checked full file details and byte sizes using **ls -l**.
7. Checked easy-to-read file sizes (KB, MB) using **ls -lh**.
8. Checked all files (including hidden) with complete details using **ls -la**.
9. Returned directly back to the home directory using **cd ~** and verified with **pwd**.

6. Result / Output Demonstration

All commands were executed in the terminal under user ratul on machine firefly.

6.1 Session 1: Finding Our Location, Moving Folders, and Listing Files

```
ratul@firefly: ~ — bash

ratul@firefly:~$ whoami
ratul
ratul@firefly:~$ pwd
/home/ratul
ratul@firefly:~$ ls
Desktop  Documents  Downloads  lab_demo  Music  Pictures  Videos
ratul@firefly:~$ cd lab_demo
ratul@firefly:~/lab_demo$ pwd
/home/ratul/lab_demo
ratul@firefly:~/lab_demo$ ls
backups  data.csv  notes.txt  report.pdf  scripts
ratul@firefly:~/lab_demo$ ls -a
.  ..  .lab_profile  .secret.key  backups  data.csv  notes.txt  report.pdf  scripts
ratul@firefly:~/lab_demo$ _
```

Explanation of Session 1 Results:

- **whoami** and **pwd**: Showed that we are logged in as user ratul and our starting location is /home/ratul.
- **cd lab_demo**: Moved us into the lab_demo directory. The terminal prompt automatically updated from ~ to ~/lab_demo.
- **ls** vs **ls -a**: The normal **ls** command only showed visible files. The **ls -a** command revealed hidden files like .lab_profile and .secret.key, along with . (current folder) and .. (parent folder).

6.2 Session 2: Inspecting File Details, Easy Sizes, and Returning Home

ratul@firefly: ~/lab_demo — bash

```
ratul@firefly:~/lab_demo$ ls -l
total 23568
drwxr-xr-x 2 ratul users     4096 Oct  8 10:15 backups
-rw-r--r-- 1 ratul users 14680064 Oct  8 10:20 data.csv
-rw-r--r-- 1 ratul users     1433 Oct  8 10:22 notes.txt
-rw-r--r-- 1 ratul users   262144 Oct  8 10:25 report.pdf
drwxr-xr-x 2 ratul users     4096 Oct  8 10:18 scripts

ratul@firefly:~/lab_demo$ ls -lh
total 24M
drwxr-xr-x 2 ratul users 4.0K Oct  8 10:15 backups
-rw-r--r-- 1 ratul users  14M Oct  8 10:20 data.csv
-rw-r--r-- 1 ratul users 1.4K Oct  8 10:22 notes.txt
-rw-r--r-- 1 ratul users 256K Oct  8 10:25 report.pdf
drwxr-xr-x 2 ratul users 4.0K Oct  8 10:18 scripts

ratul@firefly:~/lab_demo$ ls -la
total 23580
drwxr-xr-x 4 ratul users     4096 Oct  8 10:28 .
drwx------ 9 ratul users     4096 Oct  8 09:50 ..
-rw-r--r-- 1 ratul users      342 Oct  8 10:12 .lab_profile
-rw------- 1 ratul users       64 Oct  8 10:14 .secret.key
drwxr-xr-x 2 ratul users     4096 Oct  8 10:15 backups
-rw-r--r-- 1 ratul users 14680064 Oct  8 10:20 data.csv
-rw-r--r-- 1 ratul users     1433 Oct  8 10:22 notes.txt
-rw-r--r-- 1 ratul users   262144 Oct  8 10:25 report.pdf
drwxr-xr-x 2 ratul users     4096 Oct  8 10:18 scripts

ratul@firefly:~/lab_demo$ cd ~
ratul@firefly:~$ pwd
/home/ratul
ratul@firefly:~$ _
```

Explanation of Session 2 Results:

- **`ls -l` vs `ls -lh`:** `ls -l` gives the exact file size in bytes (like `14680064` bytes for `data.csv`). Adding **`-h`** turns that big number into **`14M`**, making it very easy for humans to read.
- **`ls -la`:** Shows complete details for every file, including hidden ones. For example, we can see that `.secret.key` has strict permissions (`-rw-------`), meaning only `ratul` can read it.
- **`cd ~`:** Instantly brought us back to our home directory (`/home/ratul`) in one quick step.

7. Conclusion

In this lab, we successfully practiced the most important basic Linux terminal commands. We learned:

1. How to find our current location with **`pwd`** and move between folders using **`cd`** and **`cd ~`**.
2. How to use flags with **`ls`** to view file sizes easily (**`-lh`**) and uncover hidden files (**`-a`** and **`-la`**).
3. How Linux shows file permissions, ownership, and last modified dates.

These commands make working in the Linux terminal fast, simple, and comfortable.